Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

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Abstract

A printed claim that the scorecaster breaks online conformal coverage meets a refutation sharing its first and last authors; this paper joins them and locates the edge of the set that refutation relies on. Conformal PID states long-run coverage for any bounded scorecaster; a corollary keeps it for predictable perturbations of the deployed threshold inside an admissible radius. The claim was acted on; the readout's placement and the corollary's derivation are credited by name. At the null scorecaster and the minimal admissible saturator, that radius is *tight*. A legal dead band on the *completed* threshold exits that set at its radius; past it miscoverage is 1.000000 against $\alpha = 0.1$: coverage fails outright, not merely its rate. The edge belongs to the saturator–scorecaster pair, not the readout; elsewhere that radius is sufficient only. Under the equally legal $\hat{q} \equiv +b/2$ the failing band covers, as measured; at $\hat{q} \equiv -b/2$ both it and partial adjustment at $w = 0.999$ miscover.

1 Introduction

A refereed paper states that a scorecaster breaks Conformal PID's coverage guarantee. A later one, by the same first and last authors, proves in the same notation that a bounded perturbation of the deployed threshold does not. The later cites the earlier, so this is ordinary self-correction. The joining is what this paper adds.

An online conformal method [Angelopoulos and Bates, 2023] moves a threshold each round. Forecast stability names, scores, and prices that movement [Tunc et al., 2013, Godahewa et al., 2025, Van Belle et al., 2023, Pritularga and Kourentzes, 2024, Genov et al., 2026, Van Belle et al., 2026]. Varying temporal structure at fixed level came earlier, in forecast verification, matching predictive marginals on real producers [Gneiting et al., 2007, Pinson and Girard, 2012, Worsnop et al., 2018]. Coverage and mean length under-describe a deployed interval [Min et al., 2026, Vaze, 2026]. None of it is claimed here.

The claim and the refutation in print. Conformal PID [Angelopoulos et al., 2023] adds a *scorecaster* $\hat{q}_{t+1}$ to a saturating error integrator, asking only that it be predictable and lie in $[-b/2, b/2]$: "any function of the past" (Theorem 1). Anything placed in that slot is already quantified over. The claim reads "[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical coverage guarantee" [Li and Rodríguez, 2025]. It is stated and acted on in that paper's baselines paragraph, verbatim on refereed printed page 18443. It is refuted twice, with proof. Li et al. [2025, Corollary 2] add to the deployed threshold any predictable d_t with $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ and $\mu_t \in [0, 1]$ predictable. Since $\hat{q}_t + d_t$ still lies in $[-b/2, b/2]$ it is itself a legal scorecaster, so the result "has exactly the CPID error-integration form" and $|E_T| = o(T)$. The same generic-slot argument carries AcMCP's long-run coverage [Wang and Hyndman, 2024]. Both identifiers and "scorecaster" were searched across arXiv full text, arXiv abstracts, OpenReview, and Semantic Scholar's six citers of Li and Rodríguez [2025]: none returns both the claim and the corollary. Hu et al. [2026] still call scorecaster choice "arbitrary" and lacking "principled guidance" while asserting valid coverage: their complaint is model selection, not validity. **We claim neither the placement nor the derivation.**

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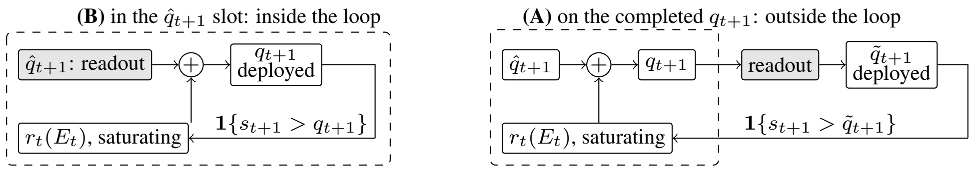

Figure 1: One scalar readout (grey) in its two places in (1); dashed is the loop r_t closes. In **(B)** the loop closes on the integrator's own output, and both readouts keep $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster's range, inside Corollary 2's admissible set. In **(A)** it closes on $\tilde{q}_{t+1}$, which it never computed, and nothing keeps the deployed sequence inside that set.

Where Corollary 2 stops, its radius condition binds. It is a *retention* result, silent outside that radius, and it leaves open whether the radius is sharp. A legal perturbation leaves it: the L_1 dead-band family characterised here on the *completed* threshold, whose edge coincides with the corollary's radius at the null scorecaster (Section 3). Past it coverage fails outright: the published sufficient condition is necessary there. Appendix C sets both claims beside four vocabularies for the same movement.

2 Setup

The producer. A forecaster fit before t induces a score s_t, and a threshold q_t is emitted before y_t is revealed. The deployed set is $C_t = \{y : s_t(x_t, y) \leq q_t\}$ and $\mathrm{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$. The target is long-run coverage, $T^{-1} \sum_{t \leq T} \mathrm{err}_t = \alpha + o(1)$, with no model on the data. Conformal PID [Angelopoulos et al., 2023] manipulates the threshold on the score scale:

$$q_{t+1} \;=\; \hat{q}_{t+1} \;+\; r_t\Big(\textstyle\sum_{i \leq t} (\mathrm{err}_i - \alpha) \Big), \tag{1}$$

Angelopoulos et al.'s iteration (5). Its r_t saturates: their condition (4) requires $|r_t(x)| \geq b$ with the sign of x once $|x| \geq c\, h(t)$, for h nonnegative, nondecreasing, and sublinear. With the accumulator $E_t = \sum_{i \leq t} (\mathrm{err}_i - \alpha)$, Proposition 2 gives $|E_T| \leq c\, h(T) + 1$, read on realised scores.

Two readouts, two placements. Damping a sequentially updated scalar has two standard readouts, both prior work. A quadratic cost gives linear partial adjustment $\tilde{q}_{t+1} = (1-w)\hat{q}^{\mathrm{raw}}_{t+1} + w\tilde{q}_t$ [Gârleanu and Pedersen, 2013], published as post-processing [Godahewa et al., 2025] and applied to a deployed threshold [Binny and Dixit, 2025, Eq. 13]. A proportional cost gives a no-trade band, which moves only when the target is more than τ away and then pulls back exactly τ [Constantinides, 1986, Davis and Norman, 1990]. Either readout has two positions in (1), drawn in Figure 1; **(A)**, Placement A, is measured here. *The dead band (L_1) is not a stylistic alternative to the partial-adjustment (L_2) form*; it is the family whose failure Section 3 characterises. Neither is safe by construction: L_2 forfeits Proposition 2's rate while covering, L_1 past the radius loses coverage, and at the legal $\hat{q} \equiv -b/2$ both lose it.

The quantity, and the names it already has. We report the deployed threshold's *travel*, $\mathcal{T}_H = \sum_{t=2}^{H} |q_t - q_{t-1}|$, named after actuator travel in process control. The arithmetic is not new, and the name is a convenience rather than a contribution, used here for one reading only: Appendix C names ten for the same movement across four vocabularies.

Harness. The harness fixes $\alpha = 0.1$ and $b = 2$. Its saturator $r_t(x) = b\,\mathrm{clip}(x/(c\,h(t)), -1, 1)$, with $c = 1$ and $h(t) = \log(t+2)$, meets (4) with equality. The scorecaster is null and legal, reducing (1) to the pure error integration Proposition 2 addresses, so the readout is the only manipulated variable. Neither choice is neutral: condition (4) is a lower bound met with equality here, and Theorem 1 permits scorecasters $b/2$ either side of null, predictable but not required to be constant in t. The adversary is specified, not tie-broken. It plays $s_t = \mathrm{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ with $\varepsilon = 10^{-9}$, against the *deployed* $\tilde{q}_t$: the $\varepsilon \downarrow 0$ reading of "sets the score at the threshold". That play fires the strict indicator and is legal on both sides of Theorem 1's $[-b/2, b/2]$, so $b/2 + b/2 = b$ is tight. *That specification names one member of a class*: any adapted $s_t \in [-b/2, b/2]$ is legal, the constant $s_t \equiv -b/2$ included. Every number in Table 1 (App. A) is measured against the one adversary above; Section 3's two boundaries, which differ from each other, are quantified over the whole class.

3 The edge of the admissible set

The published retention result. The radius restated in Section 1 is the one Li et al. [2025, Corollary 2] prove, and it *retains* $|E_T| \leq c\,h(T) + 1$. A downstream partial adjustment of weight w leaves it. Table 1 (App. A) prices that departure: five of the table's eleven arms exceed the bound at $T = 10^6$. The rate forfeit is a property of the smoothing *strength*, not Placement A, and it is bought: travel falls from 28,311 to 202. A readout on the completed q_{t+1} touches neither r_t nor the integrator's argument, so it cannot put their condition (4) at risk. It delays a correction already computed, so smoothing the output feeds E_t rather than damping it.

The L_1 dead band and the four boundaries. The dead band of Section 2 keeps $|\tilde{q}_{t+1} - q_{t+1}| \leq \tau$ at every round, so "inside Corollary 2" reads $\tau \leq \mu_t(b/2 - |\hat{q}_t|)$. Under saturation the deployed value is pinned τ below the integrator's ceiling. Outside the radius two levels decide the answer and condition (4) names only one of them. Write $A_t^+ = \sup_x r_t(x)$ and $A_t^- = \inf_x r_t(x)$ for the integrator's extremes, and $\Lambda_t^+ = \inf_{x \geq c\,h(t)} r_t(x)$, $\Lambda_t^- = \sup_{x \leq -c\,h(t)} r_t(x)$ for the levels it is *guaranteed* to reach once the accumulator clears the radius. Condition (4) says exactly $\Lambda_t^+ \geq b$ and $\Lambda_t^- \leq -b$. That $\Lambda_t^\pm = A_t^\pm$, i.e. that the integrator reaches its extremes where (4) puts them, is a further hypothesis, and not one (4) supplies. Set

$$\tau^{\star\pm} \;=\; \sup_t \big(|A_t^\pm| \pm \hat{q}_{t+1}\big) - b/2, \qquad \sigma^\pm \;=\; \inf_t \big(|\Lambda_t^\pm| \pm \hat{q}_{t+1}\big) - b/2. \tag{2}$$

Failure, over the whole admissible class. If $\tau > \tau^{\star+}$ then $\hat{q}_{t+1} + A_t^+ - \tau < b/2$ at every round, so $\tilde{q}_{t+1} \leq \max(\tilde{q}_t, q_{t+1} - \tau)$ makes $\{\tilde{q} < b/2\}$ absorbing: a release upward lands below $b/2$, and a hold or a release downward cannot lift the deployed value at all. Against $s_t = \mathrm{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ the indicator is then $\mathbf{1}\{\tilde{q}_t < b/2\}$ and every round misses. If $\tau \geq \tau^{\star-}$ the constant legal play $s_t \equiv -b/2$ makes $\{\tilde{q} \geq -b/2\}$ absorbing by the mirror inequality, so no round can miss and the realised rate is 0 rather than α. Either way long-run coverage is lost, with $|E_T|$ linear in T. Nothing here asks $\hat{q}$ to be constant, asks the extremes to be attained anywhere, or asks $A_t^\pm$ to be finite beyond what the hypothesis already forces. **Retention, keyed to the reach rather than the supremum.** If $\tau \leq \sigma^+$ then $E_{t-1} \geq c\,h(t-1)$ forces $q_t \geq b/2 + \tau$ and so $\tilde{q}_t \geq b/2$, whence $\mathrm{err}_t = 0$ against *every* legal adversary; if $\tau < \sigma^-$ the mirror forces $\mathrm{err}_t = 1$ below $-c\,h(t-1)$. The accumulator cannot then leave $[-c\,h(t), c\,h(t)]$ by more than one round, and $|E_T| \leq c\,h(T) + 1$: Proposition 2's own bound, retained verbatim rather than weakened to $o(T)$. Using condition (4) alone this already certifies $\tau \leq b/2 + \inf_t \hat{q}_t$ and $\tau < b/2 - \sup_t \hat{q}_t$, which for a constant scorecaster is Corollary 2's radius at $\mu_t = 1$.

The two directions meet, with no gap, on the sub-class the five settings inhabit. If $\hat{q}_t \equiv \hat{q}$, if $\Lambda_t^\pm = A_t^\pm$, and if neither depends on t, then $\sigma^\pm = \tau^{\star\pm}$ and **coverage is retained against every legal adapted adversary if and only if $\tau \leq \tau^{\star+}$ and $\tau < \tau^{\star-}$.** For a symmetric saturator that is one line: coverage if and only if $\tau \leq \sup_x r_t(x) - b/2 - |\hat{q}|$, with the endpoint included only when $\hat{q} < 0$. Every setting run here is in that sub-class, so the sharp form applies to all five. **What was printed before is the failure half.** $\tau^{\star+}$ is the rule Figures 2 and 3 (App. A) draw, and all 19 dead-band runs agree with it read against the adversary the harness plays. It is not a retention condition, it omits $\tau^{\star-}$ entirely, and Table 2 (App. A) separates the two with legal objects only. The supremum can sit where the dynamics never go: since $|E_t| \leq (1-\alpha)t$, a saturator equal to the harness's inside $|x| \leq t^2$ and to $\pm 10b$ outside it satisfies (4), reads $\tau^{\star+} = 19$ and fails at $\tau = 1.001$; and a legal time-varying $\hat{q}$ splits $\sup_t$ from $\inf_t$, $\hat{q}_t = b/2$ on t a power of two and 0 elsewhere reading $\tau^{\star+} = 2$ against $\sigma^+ = 1$.

The sharp form is one number, and its hypotheses are what buy it. Specialise to a symmetric saturator that meets (4) with equality and attains its extremes there ($\Lambda_t^\pm = A_t^\pm = \pm b$; the harness's $r_t = b\,\mathrm{clip}(x/(c\,h(t)), \pm 1)$ is one, Section 2), and to a constant $\hat{q}$. Then

$$\min(\tau^{\star+}, \tau^{\star-}) \;=\; b/2 - |\hat{q}|, \tag{3}$$

with the endpoint admissible only when $\hat{q} < 0$: exactly Corollary 2's radius at $\mu_t = 1$, now as an if-and-only-if. **The necessity direction above assumes none of it**: $\tau > \tau^{\star+}$, or $\tau \geq \tau^{\star-}$, forfeits coverage against every admissible r_t and every legal predictable $\{\hat{q}_t\}$, extremes attained or not, and that unrestricted statement is what this section claims. The two hypotheses are load-bearing rather than decorative: the unattained supremum and the varying-$\hat{q}$ split two paragraphs above are exactly what they rule out. Equation (3) is where the reach meets the supremum, not evidence that they always do.

The mirror boundary on this paper's grid. The τ-cliff run's exact-threshold column, read against the mirror adversary $s_t \equiv -b/2$ (Figure 2, App. A), locates $\tau^{\star-} = 1$ at the null scorecaster, closed on the failing side, where (2) puts it. Both 1.000000 and 0.000000 are failures of the same order, $|E_T| = (1-\alpha)T$ and αT, **so the failure criterion is $|E_T|$, not the miscoverage rate**. Once $\hat{q}$ varies, the miscoverage rate no longer indicates whether coverage has failed.

Both degenerate edges, and what they need. An unbounded supremum is not on its own enough for $\tau^{\star+} = \infty$ to mean what it looks like: the same unreachable-set construction with $r_t(x) = x$ past $|x| > t^2$ has no finite supremum and fails at $\tau = 1.5$. What the tangent family has, and that construction does not, is a genuine reach: $\Lambda_t(X) = \tan(\arctan(b)X/(c\,h(t)))$, so the accumulator level at which the deployed threshold is forced to $b/2$ is $\kappa(\tau)\,c\,h(t)$ with $\kappa(\tau) = \arctan(b/2 + \tau - \hat{q})/\arctan(b)$, finite at every finite τ. Coverage is retained at every width, as measured; but $\kappa(\tau) > 1$ once $\tau > b/2 + \hat{q}$, so Proposition 2's own *constant* is forfeited past $\tau = b/2$ while coverage is not. At $\tau = 1.5$, $T = 10^6$ the excursion is 15.10 against Proposition 2's 14.8155 and against $\kappa(1.5)\,c\,h(T) + 1 = 15.8530$. At the other edge, $\sup_t \hat{q}_t = -b/2$ forces $\hat{q} \equiv -b/2$, since every $\hat{q}_t$ is at least $-b/2$ already. There $q_{t+1} - \tau \le -b/2 + b - \tau < b/2$ for every $\tau > 0$ and every t, and $\tilde{q}_1 = 0 < b/2$, so the failing set is absorbing from round 1 with no transient: the admissible window is the single point $\tau = 0$, and $\tau = 0.5$ returns 1.000000, as does partial adjustment at $w = 0.999$.

Tightness of the published condition. At $\mu_t = 1$, $\hat{q} \equiv 0$, $\tau = 0.9$ covers at 0.100012 while $\tau = 1.5$ returns 1.000000 with $\max_t |E_t| = 900{,}000$, 60,747 times Proposition 2's bound at $T = 10^6$ (Table 1, App. A); (2) makes that an if-and-only-if against the whole legal adversary class, not a pattern across five settings. **The claim is that necessity: a tightness result about someone else's sufficient condition**, smaller and more precise than a claim that post-processing forfeits the rate. Elsewhere it stays sufficient only, and provably so: the tangent integrator above forfeits no coverage at any width Corollary 2 permits. At $\hat{q} \equiv +b/2$ the looseness goes the other way and disappears: Corollary 2 permits radius 0, and (2) permits no width at all against the mirror adversary, the unsmoothed control included. Against the specified adversary $\tau = 1.5$ covers there. That asymmetry makes "tight" a located claim: a tie-convention property of the cited closed statements.

The boundary read as a bet. The boundary has a betting reading (App. B): under a Bernoulli(α) null on err_t, imposed for that reading alone and on nothing else here, an anytime-valid test convicts the $\tau = 1.5$ arm past the edge at the 5% level and cannot reject inside it at any horizon. It adds a diagnostic and a reading, not a result. No result above assumes that null or is certified by that test.

4 Limitations

The retention direction is proved with no gap only where the two boundaries meet: a constant scorecaster and a saturator attaining its extremes at the condition-(4) radius, which is every setting run here; a genuinely time-varying $\hat{q}$ leaves the band $\sigma^+ < \tau \le \tau^{\star+}$ open in neither direction (Section 3). And the excursion is specific to the saturator's *level*, which condition (4) bounds only from below. **The inherited guarantee is weak where it applies.** Under the tangent integrator $h(t) = t/\log t$, the certified gap $(c\,h(T) + 1)/T$ decays as $O(1/\log T)$ rather than $O(1/T)$. The forfeited certificate was not a tight one. **The dead-band widths are absolute, and the failure is total only against the adversary.** τ is in score units with $b = 2$, so the failing $\tau = 1.5$ is $0.75\,b$. Under i.i.d. scores it returns 0.249376 at $T = 10^6$, an adversary-free confirmation of the mechanism rather than a retreat. The threshold pins at $b - \tau = 0.5$, and uniform scores on $[-b/2, b/2]$ put $P(s > 0.5) = 0.25$ exactly. **The harness was rebuilt against numbers already read**, and that carries less weight than a rerun done without sight of them. **Placement B is conceded, not tested. The i.i.d. check above used one seed** (seed 20260820, $T = 10^6$). **The primary, adversarial harness uses none**, and **only its null-scorecaster edge is bracketed**; the others are one-sided. **What remains open is the band between the two boundaries.** For a genuinely time-varying $\hat{q}$, coverage in $\sigma^+ < \tau \le \tau^{\star+}$ is neither proved nor falsified; Table 2's power-of-two rows sit there, and closing that band is future work, not a sweep this paper has already run.

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Code and data

The harness, its frozen configuration, the `results/` files behind every figure and table, the generators that read them, and this project's audit trail are at `https://[REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION]`.

A Supporting measurements

Table 1: Placement A: a one-scalar readout on the *completed* threshold, under one deterministic adversary, on one pass to $T = 10^6$; all eleven arms run are listed. Proposition 2's bound is 13.2061 at $T = 2{\times}10^5$ and 14.8155 at $T = 10^6$; "ratio" is $\max_t |E_t|$ over it, and "Sat." the fraction of rounds on which (4) delivers its full $\pm b$. Every number is measured against the adversary Section 2 specifies.

Readout on the completed threshold	miscov. $2{\times}10^5$	miscov. 10^6	$\max_t \lvert E_t\rvert$ 10^6	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	0.100007	7.80	0.53	0.0000	28,311
partial adj. $w = 0.5$	0.100030	0.100007	7.70	0.52	0.0000	18,113
partial adj. $w = 0.9$	0.100030	0.100007	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	0.100006	63.00	4.25	0.0007	1,023
partial adj. $w = 0.999$	0.100035	$\mathbf{0.100004}$	623.70	42.10	0.0097	202
growing time constant $p = 0.5$	0.100030	0.100006	12.60	0.85	0.0000	330
growing time constant $p = 0.9$	0.100030	0.100024	53.00	3.58	0.3181	16
running mean	0.099940	0.100010	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	0.100005	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	0.100012	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	$\mathbf{1.000000}$	$\mathbf{1.000000}$	900,000	60,747	1.0000	0.5

Table 2: Legal $(r_t, \hat{q})$ configurations on which the printed $\tau^\star$ over-certifies, each under two legal adversaries. "Proved window" is where (2) certifies retention against every legal adversary; between it and the failure boundaries $\tau^{\star\pm}$ neither direction is claimed. The rows built on a supremum that the dynamics cannot reach, or that is never attained, sit in that gap. Here $\emptyset$ marks a configuration for which no width is certified, $\tau = 0$ included, and $T = 10^5$ except the two power-of-two rows, at $T = 10^6$. The last row carries no dead band at all.

				$s_t = \tilde{q}_t + \varepsilon$		$s_t \equiv -b/2$	
$(r_t, \hat{q})$ configuration	τ	printed $\tau^\star$	proved window	miscov.	$\max_t \lvert E_t\rvert$	miscov.	$\max_t \lvert E_t\rvert$
$\hat{q} \equiv +0.4$, level b	0.599	1.4	$\tau < 0.6$	0.100010	7.20	0.099890	11.60
$\hat{q} \equiv +0.4$, level b	0.8	1.4	$\tau < 0.6$	0.100050	8.70	$\mathbf{0.000000}$	10,000
$\sup_x r_t = 10b$, off the reach	1.001	19	$\tau < 1$	$\mathbf{1.000000}$	90,000	0.000000	10,000
$\sup_x r_t$ unbounded, off the reach	1.5	∞	$\tau < 1$	$\mathbf{1.000000}$	90,000	0.000000	10,000
$A^+ = 4b$, $A^- = -b$	1.5	7	$\tau < 1$	0.100020	4.30	$\mathbf{0.000000}$	10,000
level $4b$, $\hat{q} \equiv 0$	6.999	7	$\tau < 7$	0.100070	12.20	0.100050	11.60
level $4b$, $\hat{q} \equiv 0$	7.0	7	$\tau < 7$	0.100090	12.20	$\mathbf{0.000000}$	10,000
$\sup_x r_t = 2b$, unattained	2.99	3	$\tau < 1$	0.100320	211.10	0.098010	210.50
$\sup_x r_t = 2b$, unattained	3.0	3	$\tau < 1$	$\mathbf{1.000000}$	90,000	0.000000	10,000
$\sup_x r_t = 2b$, attained	3.0	3	$\tau < 3$	0.100000	12.20	$\mathbf{0.000000}$	10,000
$\hat{q}_t = b/2$ on powers of 2	1.0	2	$\emptyset$	0.100009	14.30	$\mathbf{0.000000}$	100,000
$\hat{q}_t = b/2$ on powers of 2	1.9	2	$\emptyset$	$\mathbf{0.110960}$	84,745.70	0.000000	100,000
$\hat{q} \equiv +b/2$, no band	0.0	2	$\emptyset$	0.100000	0.90	$\mathbf{0.000000}$	10,000

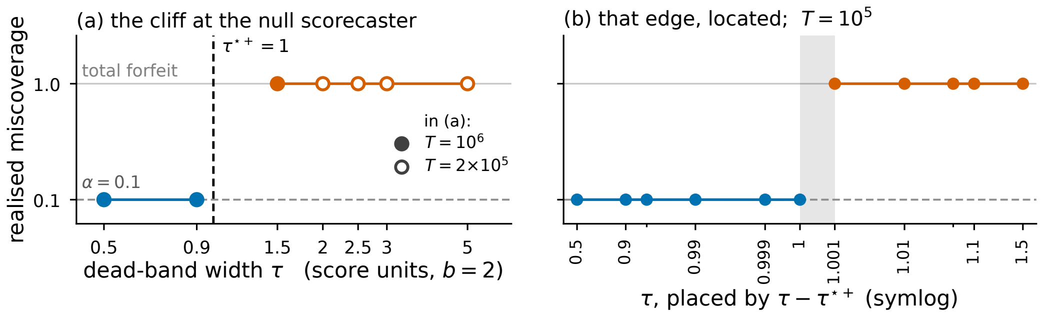

Figure 2: Realised miscoverage against dead-band width τ at the null scorecaster, $\hat{q} \equiv 0$ with a saturator of level b, under the *specified* adversary, where the upward boundary $\tau^{\star+}$ of (2) is 1. **(a)** The full sweep. Filled markers run to $T = 10^6$ and the four widest bands, drawn open, to $T = 2\times10^5$; blue covers and orange forfeits. The dashed rule is at $\tau^{\star+}$ and the grey band spans the widest covering width to the narrowest failing one. **(b)** The same setting at $T = 10^5$ over eleven measured widths, placed by $\tau - \tau^{\star+}$ on a symlog axis that magnifies the neighbourhood of $\tau^{\star+}$, so visual spacing there is not proportional to true distance in τ. The linear zone is $|\tau - \tau^{\star+}| \leq 10^{-3}$, with unlabelled minor ticks at $\tau = 0.95$ and 1.05. Figure 3 gives the other four $(r_t, \hat{q})$ settings.

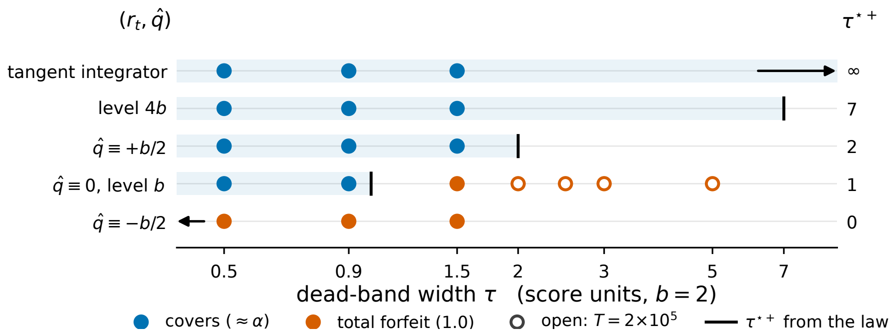

Figure 3: All 19 dead-band runs at the five admissible $(r_t, \hat{q})$ settings, rows ordered by $\tau^\star$ on the shared horizontal τ axis: blue markers cover at $\approx \alpha$ and orange markers return 1.000000, filled to $T = 10^6$ and open to $T = 2\times10^5$, all under the *specified* adversary. The rule on each row is that setting's upward boundary $\tau^{\star+} = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, printed at the right, with the tinted band $\tau \leq \tau^{\star+}$; two rules fall off the axis and are drawn as arrows, $\tau^{\star+} = 0$ at $\hat{q} \equiv -b/2$ and $\tau^{\star+}$ unbounded under Angelopoulos et al.'s tangent integrator. The mirror boundary $\tau^{\star-}$ of (2) is not drawn here and differs from $\tau^{\star+}$ at the two non-null scorecasters.

B The betting reading

The boundary is stated in $|E_T|$, which is the net gain of a Skeptic who pays α each round for err_t; that is what gives it a betting reading. The reading Section 3 points at is Ville's inequality applied to the standard martingale $M_t(\lambda) = \prod_{i \leq t} \big(1 + \lambda(\mathrm{err}_i - \alpha)\big)$, whose logarithm depends on the path only through t and E_t [Shafer and Vovk, 2019, Ramdas et al., 2023, Vovk and Wang, 2021], under the imposed null that $\mathrm{err}_t \mid \mathcal{F}_{t-1}$ is Bernoulli(α), with $\mathcal{F}_{t-1}$ the history through round $t-1$. Ville's inequality needs each bet to be predictable, fixed by the history before the round it is scored on, so the direction of failure cannot be read off the realised path first. The diagnostic is therefore the two-sided mixture $M_t = \frac{1}{2}M_t(+1) + \frac{1}{2}M_t(-1)$, which is a test martingale for the same null because a convex combination of test martingales is one, and which fixes no direction in advance; both signs lie in $[-1/(1-\alpha),\, 1/\alpha]$, the range on which $M_t(\lambda) \geq 0$ on every path, the magnitude 1 is the same for every arm and both directions, and the mixture costs a factor of two against a correctly guessed direction. So read, an anytime-valid test of the deployed intervals' calibration convicts the $\tau = 1.5$ arm on its sixth interval at the 5% level, where $M_5 = 12.38$ and $M_6 = 23.52$ against the rejection threshold $1/0.05 = 20$. Inside the edge the same test cannot reject at that level at any horizon: it peaks at 2.03 for $\tau = 0.5$ and 3.92 for $\tau = 0.9$, and from round 62 on Proposition 2's bound on $|E_t|$ forbids rejection on every retained trajectory. M_t is an e-value for that imposed null alone; the accumulator itself is signed and is neither an e-value nor an e-process, and no result in Section 3 assumes that null or is certified by that test.

C Related work

One quantity, four vocabularies. Four literatures name the movement of a sequentially updated forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC, MQC/SMQC, congruence, nervousness); *jumpiness* [Zsótér et al., 2009], *convergence index* [Ehret, 2010], *(in)consistency* [Pappenberger et al., 2011]; and *switching cost*. The same arithmetic is the Multi-Quantile Change over a predictive distribution [Zanotti, 2026, v3, §3.4.1], IACER in applied control [Sarhadi, 2025], and a path length in dynamic-regret online learning. Online conformal prediction gives distribution-free long-run coverage for any bounded scorecaster [Angelopoulos et al., 2023, Thm. 1, Prop. 2]. It generalises the running-error feedback that adaptive conformal inference [Gibbs and Candès, 2021] and its distribution-shift variant DtACI [Gibbs and Candès, 2024] already apply directly to the threshold. Forecast stability treats the movement as a functional. It is read out as post-processing [Godahewa et al., 2025, Eq. 3] and priced against a decision to switch [Genov et al., 2026, §4.2], [Van Belle et al., 2026].

The object that settles a deployed interval's validity (the admissible set around the deployed threshold) sits in the first of those four literatures. They are where this paper's claim and its concessions have to be placed. **What is conceded, by name.** The placement is prior work: Angelopoulos et al. [2023, Thm. 1, Prop. 2] state coverage, with a rate, for any admissible integrator and any scorecaster, and they run a smoothing-family model in that slot themselves. This paper proves that admissible set tight. Dupuy et al. [2026, Appendix A] give the generic argument, and Duerst et al. [2024] adopt that scorecaster. The derivation is prior work too. That concession matters most. Li et al. [2025, Corollary 2] prove the retention statement in Conformal PID's own notation, with Wang and Hyndman [2024] an independent second instance. Section 2 concedes the smoother object, the metric arithmetic, and both readout maps. Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a smoother inside their own validity argument. In one more vocabulary, this is integrator anti-windup with a feedforward path: free-until-the-bound-binds is its defining specification, not a finding [Galeani et al., 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model* update on the conformal score. Their one non-standard assumption is that $\sum_{t \le T} q_t \le O(\alpha\, h(T))$ for a sublinear h. They support it with a plot rather than deriving it. Verification's matched-level design tests real producers from 2012, with a control stronger than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]. Switching-cost online learning adds a trade-off: sublinear regret is bought by growing θ with T, and growing θ grows the ratio [Andrew et al., 2013, Thm. 7].

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   Justification: Section 3 states the full admissible class (Conformal PID's condition (4) and Theorem 1's bound), defines the two boundary quantities $\tau^{\star\pm}$ and $\sigma^{\pm}$ in (2), and proves both the failure direction (over the whole admissible class, no attainment or constancy assumed) and the retention direction (keyed to the guaranteed reach), showing the two meet with no gap on the sub-class every tested setting inhabits; the closed form (3) is stated as an if-and-only-if with its two enabling hypotheses named explicitly as load-bearing rather than decorative. **One honest caveat**: the proof is presented in narrative prose within the section text, with assumptions stated in the surrounding sentences and conditions given as labelled equations, rather than as a numbered `theorem`/`proof` LaTeX environment with a formal statement–proof separation. The mathematical content (assumptions, both proof directions, both degenerate edge cases) is complete and was independently re-derived and adversarially attacked in this project's own process; converting the presentation to a numbered theorem environment is a formatting change deferred to a future revision, not a gap in the argument.

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Answer: [No]

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